

XIANGYU ZENG

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🔗 <https://lanxingxuan.github.io/> · Google Scholar Link

EDUCATION BACKGROUND

Shandong University

Sep. 2020 – Jun. 2024

B.Eng. School of Computer Science and Technology

Course GPA: 92.1/100 Class Rank: 3/52 Comprehensive GPA: 102.1/110 Comprehensive Rank: 1/52

- *Research Team: Intelligent Media Research Center (ILearn)*
- *Honors and Awards: National Scholarship (2022), Shandong University Dean's Scholarship (2023), First-Class Academic Scholarship (2022, 2023), and Outstanding Graduate of Shandong University (2024)*

Nanjing University

Sep. 2024 – Present

Ph.D. Candidate School of Computer Science and Technology

- *Research Team: Multimedia Computing Group (MCG)*
- *Honors and Awards: National Scholarship (2025), Nanjing University Presidential Special Scholarship (2024), First-Class Academic Scholarship (2024, 2025)*

COMPETITION EXPERIENCE

- 23rd CCF CSP Certification: 370 points (Top 1%) Sep. 19, 2021
- ACM ICPC Asia Regional Contest: Silver Medal Nov. 14, 2021
- CCSP (East China): Bronze Medal Dec. 13, 2021
- ACM ICPC Asia Regional Contest: Bronze Medal Apr. 3, 2022
- CUMCM (Shandong): First Prize Sep. 18, 2022

INTERNSHIP EXPERIENCE

Shanghai AI Laboratory

Jan. 2024 – Apr. 2026

Research Intern OpenGVLab

- *Focused on pretraining and fine-tuning of multimodal video foundation models, achieved progress in long video temporal modeling and contributed to a paper accepted by ICLR 2025; participated in the Intern-Video2.5 project.*

Huawei

Jan. 2025 – Dec. 2025

Collaborative Project Noah's Ark Lab

- *Worked on online video understanding, with emphasis on memory mechanisms and real-time perception for applications such as autonomous driving; contributed to a paper accepted by NeurIPS 2025 Spotlight.*

PUBLICATIONS

Selected Works

- **Video-o3: Native Interleaved Clue Seeking for Long Video Multi-Hop Reasoning**
 - Accepted by **ICML 2026 (CCF-A)**, first author (co-first rank1)
 - Proposed Video-o3, a pioneering framework for long-video multi-hop reasoning that leverages native interleaved tool invocation.
 - Designed a Task-Decoupled Attention Masking to mitigate task interference while preserving global context. Proposed a Verifiable Trajectory-Guided Reward to optimize multi-turn interactions, significantly improving contextual efficiency by balancing active exploration coverage with precise search termination.

- Engineered an automated data synthesis pipeline to construct Seeker-173K, driving the model to achieve state-of-the-art performance across comprehensive benchmarks.

- **StreamForest: Efficient Online Video Understanding with Persistent Event Memory**
 - Accepted by [NeurIPS 2025](#) (CCF-A), first author (co-first rank1), [Spotlight \(Top 3%\)](#)
 - Developed StreamForest, a novel architecture for real-time streaming video understanding, featuring a Persistent Event Memory Forest for efficient long-term memory and a Fine-grained Spatiotemporal Window for enhanced short-term perception.
 - Achieved state-of-the-art performance across multiple benchmarks while maintaining high accuracy under extreme compression.
- **TimeSuite: Improving MLLMs for Long Video Understanding via Grounded Tuning**
 - Accepted by [ICLR 2025](#) (CCF-A), first author
 - Without relying on any external expert, TimeSuite can achieve expert-level performance in grounding tasks while maintaining considerable generalization QA capability and strong zero-shot capabilities.
 - The introduction of grounding tasks enhances the comprehensive understanding of long videos. We validated the feasibility of enhancing MLLM's comprehensive capabilities by integrating expert tasks.
- **Adaptive Edge-Aware Semantic Interaction Network for Salient Object Detection in Optical Remote Sensing Images**
 - Accepted by [TGRS 2023](#) (CCF-B), first author
 - Developed AESINet, a novel edge-aware semantic interaction network for salient object detection in optical remote sensing images, incorporating LDAM for unsupervised edge enhancement, MFEM for handling varying object scales, and DSIM for robust detection in cluttered scenes.
 - Achieved superior performance over 14 state-of-the-art methods across three benchmark datasets.

Other Works

- **InternVideo2.5: Empowering Video MLLMs with Long and Rich Context Modeling**
 - [Technical Report](#)
- **VideoChat-R1: Enhancing Spatio-Temporal Perception via Reinforcement Fine-Tuning**
 - [Technical Report](#)
- **Online Video Understanding: OVBench and VideoChat-Online**
 - Accepted by [CVPR 2025](#)
- **Task Preference Optimization: Improving Multimodal Large Language Models with Vision Task Alignment**
 - Accepted by [CVPR 2025](#)
- **Make Your Training Flexible: Towards Deployment-Efficient Video Models**
 - Accepted by [ICCV 2025](#)
- **VTTs: Visual Test-Time Scaling to Reinforce Multimodal Reasoning by Iterative Perception**
 - Accepted by [NeurIPS 2025](#)
- **VideoChat-Flash: Hierarchical Compression for Long-Context Video Modeling**
 - Accepted by [ICLR 2026](#)
- **UniFlow: A Unified Pixel Flow Tokenizer for Visual Understanding and Generation**
 - Accepted by [ICLR 2026](#)
- **RIVER: Real-time Video Interaction Benchmark**
 - Accepted by [ICLR 2026](#)

- **Balancing the Experts: Unlocking LoRA-MoE for GRPO via Mechanism-Aware Rewards**
- Accepted by [ICLR 2026](#)
- **FreeRet: MLLMs as Training-Free Retrievers**
- Accepted by [ICML 2026](#)
- **ReaForest: Fostering Generative Video Reasoning for Spatial Planning**
- Accepted by [ICML 2026](#)